

A Parameter-Uniform Affine Hénon Horseshoe: Exact Coding, Stability Traces, and a Trace-Class Owner

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Abstract

We exhibit a two-parameter family of piecewise-affine Hénon horseshoes for which the hyperbolic geometry, full-shift coding, every-period fixed-point weights, and a trace-class operator identity are uniform on an explicit rectangle. For $(\lambda, \mu) \in [3, 4] \times [1/5, 1/3]$, every based binary word of length n determines exactly one periodic point and the weighted fixed-point sum is

$$\frac{2^n}{(\lambda^n - 1)(1 - \mu^n)}.$$

We realize this sum as the n th trace of a natural stability-mode operator, derive its Fredholm product, prove a uniform zero-free disk $|z| < 3/2$, and give explicit trace-norm Lipschitz constants across the whole parameter rectangle. The result closes a structural uniform-parameter gate; it does not compare the determinant with an external spectral target.

1 Progress over the prior gate

The prior Hénon-track evidence included individual parameter witnesses and finite symbolic prefixes. Such evidence does not imply persistence on a region. Here the advance is a single theorem, with explicit constants, on a two-dimensional parameter rectangle. It simultaneously controls strip geometry, hyperbolicity, coding, all periods, a global Hilbert-space owner, and its determinant. No fitted target data enter the construction.

2 The affine horseshoe

Let $\mathcal{P} = [3, 4] \times [1/5, 1/3]$. For $(\lambda, \mu) \in \mathcal{P}$, set

$$R_0 = [0, \lambda^{-1}] \times [0, 1], \quad R_1 = [1 - \lambda^{-1}, 1] \times [0, 1]$$

and define, for $e \in \{0, 1\}$,

$$F_{\lambda, \mu}(x, y) = (\lambda x - (\lambda - 1)e, \mu y + (1 - \mu)e), \quad (x, y) \in R_e. \quad (1)$$

Each vertical strip maps across the square into a horizontal strip. The domain gap is $1 - 2/\lambda \geq 1/3$ and the image gap is $1 - 2\mu \geq 1/3$. On either branch,

$$DF_{\lambda, \mu} = \text{diag}(\lambda, \mu), \quad \lambda \geq 3, \quad 0 < \mu \leq 1/3.$$

Thus the coordinate cones are invariant with uniform expansion and contraction.

Theorem 1 (uniform coding and periodic closure). *The maximal two-sided invariant set of (1) is conjugate to the full two-shift. A based word $e_0 \dots e_{n-1}$ has exactly one periodic point, whose initial coordinates are*

$$x_0 = \frac{\lambda - 1}{\lambda^n - 1} \sum_{j=0}^{n-1} \lambda^{n-1-j} e_j, \quad (2)$$

$$y_0 = \frac{1 - \mu}{1 - \mu^n} \sum_{j=0}^{n-1} \mu^{n-1-j} e_j. \quad (3)$$

Consequently $\# \text{Fix}(F^n) = 2^n$, and the primitive-cycle count is

$$p_n = \frac{1}{n} \sum_{d|n} \text{mobius}(d) 2^{n/d}.$$

Proof. Forward iteration of the first coordinate and backward determination of the stable coordinate give nested intervals whose diameters are bounded by 3^{-k} . The strip gaps make distinct itineraries disjoint. Solving the two affine closure equations gives (2)–(3); these points remain in their prescribed strips by the same nested-interval construction. There are therefore 2^n based fixed points. Möbius inversion removes repetitions and cyclic rooting. $\square$

3 A trace-class stability owner

At every $q \in \text{Fix}(F^n)$,

$$|\det(I - DF^n(q))| = (\lambda^n - 1)(1 - \mu^n).$$

We now obtain the resulting orbit sum as an actual operator trace. Let

$$\mathcal{H} = \mathbb{C}^2 \otimes \ell^2(\mathbb{N}_{\geq 1} \times \mathbb{N}_0), \quad J = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix},$$

and let $D_{\lambda,\mu}$ be diagonal with entry $\lambda^{-r}\mu^s$ at (r, s) . Define $K_{\lambda,\mu} = J \otimes D_{\lambda,\mu}$.

Theorem 2 (all-period trace identity). *The operator $K_{\lambda,\mu}$ is trace class, uniformly on $\mathcal{P}$, and*

$$\|K_{\lambda,\mu}\|_1 = \frac{2}{(\lambda - 1)(1 - \mu)} \leq \frac{3}{2}, \quad (4)$$

$$\text{Tr } K_{\lambda,\mu}^n = \frac{2^n}{(\lambda^n - 1)(1 - \mu^n)} = \sum_{q \in \text{Fix}(F^n)} \frac{1}{|\det(I - DF^n(q))|}. \quad (5)$$

Proof. The nonzero eigenvalue of J is 2. Both geometric series in

$$2 \sum_{r \geq 1} \lambda^{-r} \sum_{s \geq 0} \mu^s$$

converge uniformly, proving (4). Taking n th powers gives $\text{Tr } J^n = 2^n$ and

$$\sum_{r \geq 1} \lambda^{-nr} \sum_{s \geq 0} \mu^{ns} = \frac{1}{(\lambda^n - 1)(1 - \mu^n)},$$

which is (5). $\square$

4 Uniform determinant control

Trace class makes the following product an ordinary Fredholm determinant:

$$\det(I - zK_{\lambda,\mu}) = \prod_{r \geq 1, s \geq 0} (1 - 2z\lambda^{-r}\mu^s). \quad (6)$$

Its smallest-modulus zero is $z = \lambda/2$, so the determinant is zero-free for $|z| < \lambda/2$ and uniformly zero-free for $|z| < 3/2$. The word “open” matters: at $\lambda = 3$, the boundary point $z = 3/2$ is a zero.

Differentiating the absolutely convergent diagonal series gives

$$\|\partial_\lambda K\|_1 \leq \frac{3}{4}, \quad \|\partial_\mu K\|_1 \leq \frac{9}{4}$$

on $\mathcal{P}$. Integration along a coordinate path yields

$$\|K_{\lambda,\mu} - K_{\lambda',\mu'}\|_1 \leq \frac{3}{4}|\lambda - \lambda'| + \frac{9}{4}|\mu - \mu'|. \quad (7)$$

Equations (6)–(7) are global parameter statements, not neighboring-point numerical tests.

5 Exact certificate and hostile controls

The release evidence uses rational arithmetic on a 3×3 parameter grid, checks periods through 12, and replays representative words exactly. An independent implementation reconstructs every invariant. A separate SymPy program verifies the geometric-series and matrix identities, while sixteen semantic mutations test gaps, counts, traces, displayed periodic coordinates, scope flags, and route labels.

Three failure modes delimit the theorem. If $\lambda < 2$, the domain strips can overlap; if $\mu > 1/2$, the image strips can overlap. Replacing the stability denominator by $(\lambda\mu)^n$ is not a harmless convention: it breaks the trace identity. Finally, 2^n counts rooted words, not primitive cycles; the latter require Möbius inversion.

6 Route-A assessment and limitations

The structural uniform-parameter subgate is certified. The strict canonical tuple is nevertheless

$$(A1_WEAK, A2_FAIL, A3_FAIL, A4_FAIL).$$

A2 and A3 remain failed because there is no frozen target-facing protocol, missing/extra-zero census, or target divisor comparison. The internal zeros of (6) must not be rebranded as arithmetic zeros. No natural quantization is constructed, and Route B is not authorized. The active scope is `NO_BAD_EULER_OR_ROOT_NUMBER`.

7 Conclusion

This family supplies an explicit answer to a narrow but important Route-A question: exact orbit coding and a trace-class stability determinant can be made uniform on a genuine Hénon-style parameter region. The next separate task is target-facing validation under a future authorized protocol; it is not inferred from the present structural theorem.